



KCAC Position Statement

Fact Check on Kratom Benefits Due to 7-OH

The KCAC has fact checked the claim by the Holistic Alternative Recovery Trust that the benefits of leaf kratom are due to the creation of 7-OH in the body from the mitragynine in the products.

Claim: The Holistic Alternative Recovery Trust (HART) is claiming in testimony to State Legislatures across the country that 7-OH products are just as safe as kratom products because the mitragynine in kratom is converted into 7-OH, providing all the benefits of therapy.

Truth: The fresh or dried leaf of the *Mitragyna speciosa* tree contains many alkaloids that have various pharmacologic effects including enhancing serotonin, reducing inflammation, stimulating alpha-1 and alpha-2 adrenoceptors, and either partially stimulating or inhibiting opioid receptors. While the most abundant alkaloid in kratom, called mitragynine, can be metabolized into 7-OH in the body, only about 20% of the mitragynine consumed is converted. That means that mitragynine and its metabolites, as well as other alkaloids and their metabolites are all in the body at the same time but in differing concentrations over time. It is impossible to predict the safety of high dose 7-OH isolate products, based off data from leaf kratom.

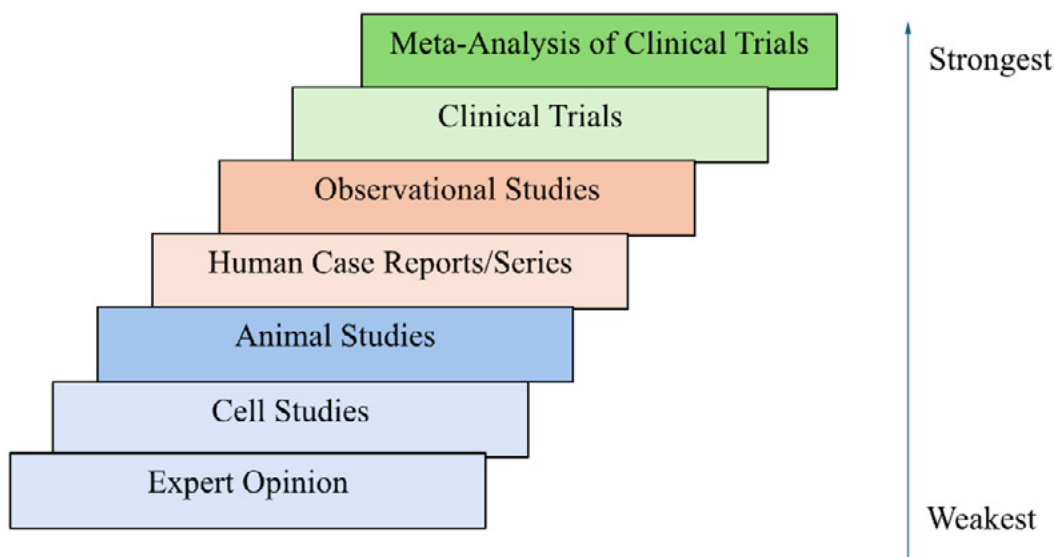
Single cell studies and animal studies are not perfect, and you cannot guarantee that what you see in animal studies will be seen in humans.^{1,2} In the hierarchy of evidence, single-cell studies and animal studies rate below even the weakest human descriptive and observational studies. Randomized and blinded human clinical trials are the strongest study evidence available and when clinical trials with low risk of bias are statistically pooled in a meta-analysis with others and show strong consistent effects without publication bias, there is the greatest level of confidence (Figure 1).²



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Figure 1. Hierarchy of evidence.



That is why clinical trials in humans are needed to truly understand the benefits and risks of any therapy. Neither kratom³ or 7-OH products have any clinical trials in people with any disease showing that they work or to fully evaluate the risks of using the products.³ In the absence of this clinical trial data, we examine normal volunteer tolerability trials, observational studies, and anecdotal evidence, which are reasonable sources of data but not the best available.² Kratom has a generally robust observational study dataset from Southeast Asia and the United States suggesting reasonable safety and efficacy as an opioid substitute, and was recently deemed well tolerated in an FDA conducted single-dose tolerability trial.³ 7-OH does not have any published observational data and only has in vitro, animal, and anecdotal (case report and descriptive) evidence.

Whole leaf kratom contains many alkaloids (mitragynine, paynantheine, speciogynine, speciociliatine, corynantheidine, mitraciliatine, isopaynantheine, and others). There is no 7-OH found in fresh *Mitragyna speciosa* leaves.³ Pharmacologic effects of kratom alkaloids in single-cell studies and animal studies include serotonin boosting, alpha-1 and 2 adrenoceptor stimulation, anti-inflammatory, muscle relaxant, and opioid receptor stimulating and blocking effects.^{3,4} Between 35% and 66% of the alkaloid fraction (total alkaloids) in whole leaf kratom is mitragynine and even this one alkaloid has a complex pharmacology. Mitragynine is a serotonin booster and alpha-1 and alpha-2 stimulator, and can indirectly and mildly stimulate opioid receptors.^{3,4} Mitragynine can partially antagonize opioid receptors when stronger opioid agonists such as morphine are present.^{3,4}

There are no pharmacokinetic studies evaluating the blood concentrations after 7-OH product consumption, so any judgement about things like the maximum blood concentration (C_{max}), the time to maximum concentration (T_{max}), or the total exposure to the drug over time (Area Under the Curve;



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AUC), is a guess. Three pharmacokinetic studies have assessed mitragynine's pharmacokinetics with doses ranging from 6.7 to 59.2 mg.⁵⁻⁷ The blood AUC and the C_{max} of 7-OH was about 20% of the parent compound mitragynine, and the T_{max} occurred around 1.7 hours after dosing for 7-OH, while the mitragynine T_{max} was more rapid at 1.0 hours (Table 1).^{5,6} In one study, they found that a specific enzyme in the intestines and the liver called CYP3A4 is responsible for conversion of mitragynine into 7-OH, and that a blocker of that enzyme called itraconazole can dramatically reduce the 7-OH created.

There is an animal study where, in Part A, they gave a high dose of mitragynine and then measured the concentrations of mitragynine and the 7-OH that was created by the breakdown of mitragynine in the brain.⁸ They measured antinociception (reduced sensation of pain) based on the rodents' response to having their tail dipped in warm water. In Part B they then directly administered isolated 7-OH to achieve the same brain 7-OH concentrations they had in Part A and measured pain-relief. They found achievement of the same 7-OH concentration in the brain, either indirectly through high mitragynine dosing or through direct administration, could achieve the same level of pain relief.⁸ However, in this animal study, the investigators used doses of 140 mg/kg of mitragynine in Phase A and only 0.7 mg/kg of 7-OH in Phase B. This mitragynine dose was 200 times that of isolated 7-OH, but the kratom products being sold to US consumers have ~10 to 40 mg of mitragynine per serving while isolated 7-OH products have ~10-20 mg of 7-OH.⁹ Clearly this animal study does not explain the pharmacologic effects of kratom, since the contribution of 7-OH would be so much smaller when the 7-OH concentrations were so much lower than when isolated 7-OH was directly administered. The antinociception model does not consider the impact of serotonin boosting on people's ability to deal with pain, the impact of reducing inflammation, or muscle relaxation on pain in humans.

This is the study HART uses to contend that all the benefits of kratom are derived from 7-OH and to suggest that isolated 7-OH has the same safety as kratom but as we just showed, it cannot prove that the benefits of kratom are solely due to 7-OH or that the risk profile is the same as kratom. This is shown elegantly in another rodent study¹⁰ where the equipotent maximal antinociceptive effect for direct mitragynine administration and direct isolated 7-OH administration from hotplate exposure was determined and then the corresponding 7-OH concentration in the brain was assessed.¹⁰ The 7-OH concentration in the brain created by the metabolism of mitragynine was 11-fold lower than the 7-OH concentration in the brain after isolated 7-OH was administered directly, even though the desensitization to the painful stimuli was the same.¹⁰ Taken together, the 7-OH created by mitragynine metabolism is one of the reasons that whole leaf kratom provides pain relief in rodent models but it is not the main one.

Figure 2 provides an overview of the alkaloids found in kratom products. Mitragynine's metabolite is called 7-OH, and 7-OH's metabolite is called mitragynine pseudoindoxyl. Neither 7-OH or mitragynine pseudoindoxyl is found in fresh kratom leaves and only trace amounts of 7-OH (less than 2% of total alkaloids) are present in dried kratom leaves. It is important to appreciate that whole leaf kratom is the sum total of all these alkaloids while 7-OH is a potent opioid receptor stimulator and its active metabolite, mitragynine pseudoindoxyl, is as well.¹¹ 7-OH has been shown to have pain reduction, tolerance, withdrawal, and addiction effects in rodent models that are similar to or greater potency than morphine, where mitragynine does not have these effects, even though some of the mitragynine is being converted into 7-OH in the body.^{3,12}

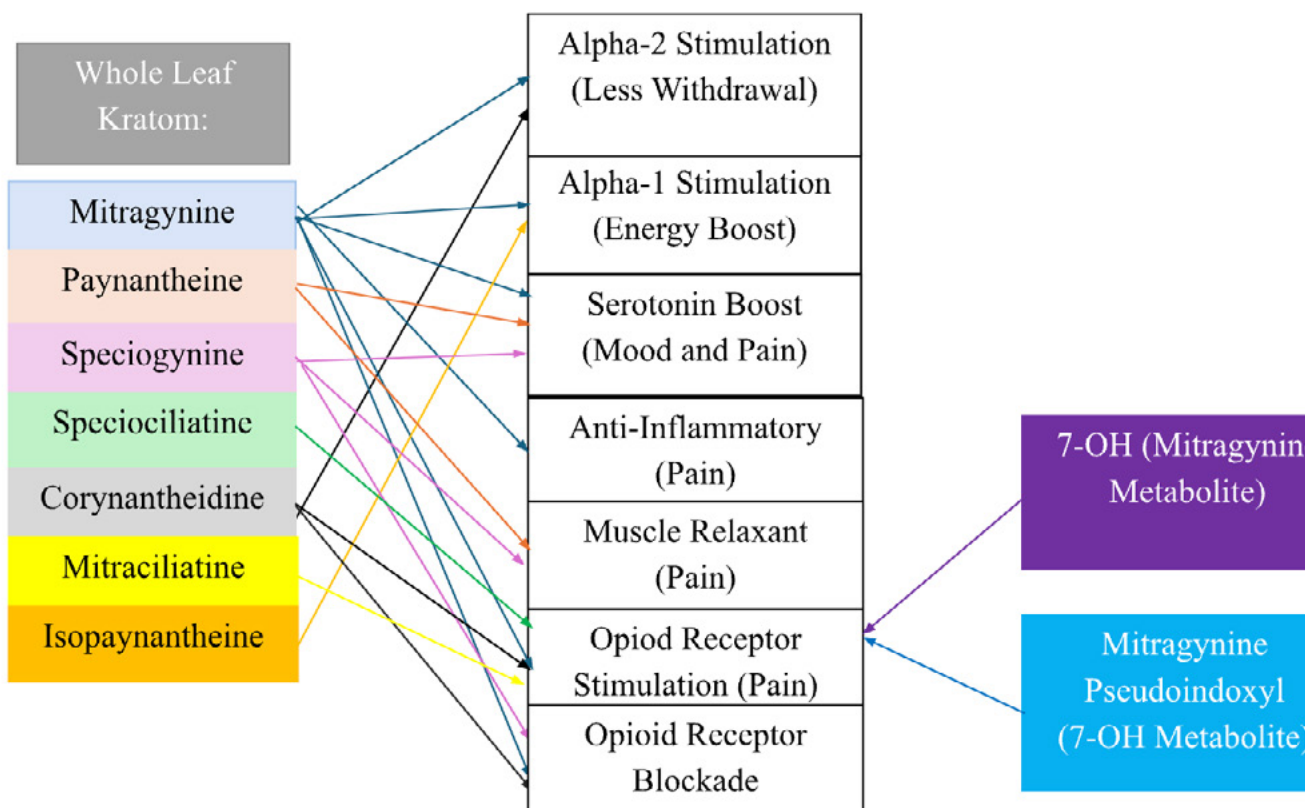


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Whole leaf kratom products and mitragynine extracts have preclinical study data to suggest a very low risk of respiratory depression, but this does not seem to be the case for 7-OH. In a recent rodent study, the use of isolated 7-OH reduced respiratory rate (the number of times a rodent breathes per minute) and the minute ventilation (the multiplication of the respiratory rate and the tidal volume (the amount of air exchanged each time a rodent breathes) in a dose related manner (high dose, greater impairment of breathing).¹³ The magnitude of these effects were similar to rodents given morphine, but mitragynine did not negatively impact either of these parameters in this rodent study. Furthermore, when naloxone was given to morphine dosed rodents the minute ventilation was strongly reversed while in isolated 7-OH dosed rodents, the reversal was modest.¹³ The similar dose related respiratory depressant effects of isolated 7-OH and morphine were found in another rodent study, while mitragynine had a ceiling effect (meaning that using higher doses did not increase the risk of more severe respiratory depression like treatment with 7-OH does).¹⁴ As such, the animal data suggests that isolated 7-OH poses a risk of precipitating the very respiratory depression that regular opioids do and the higher the dose used, the greater the risk. It is disingenuous to suggest, without clinical trial data assessing respiration effects, that isolated 7-OH provides greater safety than classical opioids. It is more honest to suggest that the effects of 7-OH are likely to mimic those of traditional opioids than kratom.

Figure 2. Effects of whole leaf kratom alkaloids versus 7-OH.





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Putting It All Together

Whole leaf kratom has been used for hundreds of years and medium quality evidence suggests that it has a reasonable risk profile but is not risk free. The effects are due to multiple alkaloids (active ingredients) working together or against each other to temper the magnitude of effects on multiple different areas of the body and receptor types. We do not have sufficient data to say anything definitive about kratom's ability to treat severe pain or opioid use disorder. Without FDA approval, real health claims should never be made for any product because it is deceptive and dangerous to public health. HART should not be making health claims about 7-OH products given the lack of evidence to back up the claims. Tricking people into trying 7-OH products under the charade that it is proven effective or safe is dangerous because animal studies and anecdotal experiences suggest 7-OH products are addictive, carry a risk for respiratory depression, and cause withdrawal when you stop taking them. 7-OH products have virtually no human data to base their claims of safety. It is disingenuous to say that 7-OH products are equally as safe as kratom or safer than prescription opioids. Whole leaf kratom is an amalgam of numerous alkaloids, not just mitragynine. Only a fraction of the mitragynine in kratom is converted to 7-OH, 7-OH is a much more potent opioid receptor stimulator than mitragynine, and 7-OH and its active metabolite mitragynine pseudoindoxyl seem to only stimulate opioid receptors to provide effects. Animal studies suggest that unlike kratom, 7-OH can suppress people's ability to breathe when taken in high doses. Until proven otherwise, it would be prudent to assume that the risk profile is much more likely to mimic traditional opioids such as morphine and fentanyl than a multi-ingredient product such as kratom.



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Table 1. Pharmacokinetic trials of the alkaloid mitragynine in whole leaf kratom and the 7-OH that is created in the body.

	Mongar 2024	Mongar 2024	Huestis 2024	Huestis 2024	Huestis 2024	Huestis 2024	Sempio 2024	Sempio 2024	Sempio 2024
Form	Whole Leaf Kratom Tea	Whole Leaf Kratom Tea + Itraconazole	Whole Leaf Kratom Powder	Whole Leaf Kratom Powder	Whole Leaf Kratom Powder	Whole Leaf Kratom Powder	Whole Leaf Kratom Tea	Whole Leaf Kratom Tea	Whole Leaf Kratom Tea
Number of Participants	16	16	12	12	12	13	1	3	6
Mitragynine Dose (mg)	23.6	23.6	6.7	13.3	26.6	53.2	9.9	29.6	59.2
Dosing Frequency	Once	Once	Once	Once	Once	Once	Once	Once	Once
Mitragynine T_{max} (h)	0.84	1.14	1.0	1.3	1.3	1.3	1.3	1.3	1.0
Mitragynine C_{max} (ng/mL)	159	245	17	32	57	125	23	82	211
Mitragynine AUC (ng*h/mL)	810	1,032	41	108	244	558	105	308	988
Mitragynine $t_{1/2}$ (h)	9	3	8.5	15.8	29.3	43.4	82	48	51
7-OH T_{max} (h)	1.77	2.56	1.2	1.5	1.8	1.7	1.7	2.0	1.7
7-OH C_{max} (ng/mL)	13	4	4	7	11	23	9	16	39
7-OH AUC (ng/mL)	107	34	10	26	48	125	20	55	219
7-OH $t_{1/2}$ (h)	7	3	2	3	3	5	6	3	13
7-OH/ Mitragynine C_{max} Ratio	0.08 (8%)	0.02 (2%)	0.24 (24%)	0.22 (22%)	19 (19%)	0.18 (18%)	0.39 (39%)	0.20 (20%)	0.18 (18%)
7-OH/ Mitragynine AUC Ratio	0.13 (13%)	0.03 (3%)	0.24 (24%)	0.24 (24%)	0.20 (20%)	0.22 (22%)	0.19 (19%)	0.18 (18%)	0.22 (22%)

Mongar P, Jaisi A, Inkviya T, et al. Effects of Itraconazole on Pharmacokinetics of Mitragynine and 7-Hydroxymitragynine in Healthy Volunteers. ACS Pharmacology & Translational Science 2024;7(3):823-833. DOI: 10.1021/acspsci.3c00335

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Sempio, C., Campos-Palomino, J., Klawitter, J. et al. Quantification of 11 kratom alkaloids including mitragynine and its main metabolites in human plasma using LC-MS/MS. Anal Bioanal Chem 417, 761–769 (2025). <https://doi.org/10.1007/s00216-024-05689-9>



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